

Lab: Biological Agency and Embodied Robot Design

Objective

Investigate how biological agents achieve autonomy through embodied cognition and morphological computation. You will analyze animal behavior as Active Inference, design a robot that exploits body-environment coupling, and compare biological and artificial agency.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Animal Behavior as Active Inference

Select an animal behavior (e.g., prey capture in frogs, web building in spiders, nest construction in termites) and analyze it as Active Inference:

1. What is the animal's generative model of its environment during this behavior?
2. What prior preferences drive the behavior (desired sensory outcomes)?
3. How do prediction errors guide the behavior in real time?
4. Where does morphological computation reduce the cognitive load?

Write your response here

Part 2: Soft Robotics and Embodiment

Design a soft robotic gripper inspired by biological grasping mechanisms:

1. Compare a rigid parallel-jaw gripper with a compliant, bio-inspired gripper (e.g., inspired by elephant trunk, octopus tentacle, or human hand).
2. How does material compliance serve as morphological computation, reducing the generative model's complexity?
3. What sensing modalities does the bio-inspired gripper need (tactile, proprioceptive, force)?

Write your response here

Part 3: Umwelt Analysis

The umwelt is an organism's subjective sensory world. Analyze umwelts for robotic design:

1. Describe the umwelt of a bat (echolocation), a pit viper (infrared sensing), and a star-nosed mole (touch).
2. For each, how does the umwelt constrain the organism's generative model?
3. Design a robot with an unconventional umwelt (e.g., primarily magnetic or vibrational sensing). How does this shape its agency?

Write your response here

Part 4: Agency Comparison Table

Compare biological and robotic agency:

Property	Biological Agent	Current Robots	Bio-Inspired AI Robot
Self-repair			
Metabolism/ energy			
Morphological adaptation			
Generative model depth			
Autonomy duration			

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

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- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).